IA2 - AP_DS- 2025/2026 - Tutorial 02

Exercise 1: Hospital Patient Data Analysis

Goal: Create a mini-project to manipulate patient data and visualize basic statistics.

Step 1: Create the project directory

Create a project folder named HospitalDataAnalysis.

Step 2: Create a virtual environment with a specific Python version

If you have multiple Python versions installed (e.g., Python 3.10, 3.11), specify the one you need.

Step 3: Activate it and install the required libraries

like: numpy pandas

Step 4: Create project structure

Organize your project into a clean structure by creating the following folders and files:

```
HospitalDataAnalysis/
├── env/                # Virtual environment folder (ignored by
Git)
├── data/               # Folder to store CSV files
│   └── patients.csv
├── notebooks/          # Jupyter notebooks
│   └── patients_analysis.ipynb
├── scripts/            # Optional Python scripts
├── .gitignore
└── README.md
```

Step 5: Load and explore the data

- Create a small CSV file patients.csv in the data/ folder with columns: PatientID, Age, Sex, BloodPressure, Cholesterol, Diagnosis
- Insert some values

Load it in the notebook using:

import pandas as pd

df = pd.read_csv("../data/patients.csv") df.head()

In [ ]:
```
# Run basic descriptive statistics using:

df.describe()

df['Sex'].value_counts()

df['Diagnosis'].value_counts()
```

In [ ]:
```
# Visualize the data : Create a Simple Plot using matplotlib

import matplotlib.pyplot as plt
plt.plot(df["Age"], df["Cholesterol"], marker="o")
plt.xlabel("Age")
plt.ylabel("Cholesterol")
plt.title("Cholesterol vs Age")
plt.show()
```

Step 6: Initialize Git and create README

- Create a GitHub account (if you don't already have one).
- Go to https://github.com and sign up.
- Create a new repository on GitHub called **HospitalDataAnalysis**.
  👉 Do not initialize it with a README, `.gitignore`, or license (we'll add them from local).
- Link your local project with the GitHub repository:
- Create a .gitignore file in your local project with the following content:

env/

pycache/

*.pyc

- Commit and push again:

Step 7: Share the project

- How can you export all the installed libraries in your virtual environment to a file so that someone else can recreate the same environment?
- Test recreating the environment on a different machine: